

Karnayana Rohith

Visakhapatnam, Andhra Pradesh, India

☎ +91 97011 28085 | ✉ karnayanarohith@gmail.com | 🔗 linkedin.com/in/rohith-karnayana | 🐙 github.com/karnayanarohith | 🌐 rohithkarnayana.vercel.app

Profile

Final-year B.Tech ECE student with a deep passion for understanding how systems work at their lowest level — how they break, and how to secure them. Drawn to cybersecurity not as a checklist of tools, but as a discipline of engineering resilience: patient, methodical, and precise. Equally comfortable analyzing kernel state logs, implementing threat mitigations, and building AI orchestration pipelines. Driven by a desire to study attack surfaces, secure system design, and the intersection of AI with offensive and defensive operations. Believes the best defense comes from truly understanding what you are protecting — down to the binary.

Education

Andhra University College of Engineering <i>Bachelor of Technology (B.Tech) in Electronics & Communication Engineering</i> CGPA: 8.4/10	Visakhapatnam, Andhra Pradesh 2023 – 2027
Sri Gayathri Junior College <i>Intermediate (MPC)</i> Percentage: 95%	Gotlam, Vizianagaram, Andhra Pradesh 2021 – 2023
ZPHS Siripuram <i>Secondary School Certificate (SSC)</i> Percentage: 85%	Siripuram, Srikakulam, Andhra Pradesh 2021

Technical Skills

Languages: C, Python, Bash, Assembly, JavaScript, SQL (PostgreSQL)

Cybersecurity: CVE Research & Exploitation, Log Analysis & Forensics, MITRE ATT&CK, Threat Triage & Incident Response, IAM

AI / ML: LangGraph (Agentic AI), Ollama (Local LLMs), Vector DBs (ChromaDB), FastAPI, PyTorch, OpenCV

Systems & OS: Embedded Linux, Android Bootloader Internals, process/memory debugging, filesystem structures

Tools & Utilities: Git, Docker, Linux CLI, GDB, ADB, Fastboot, Wireshark, Splunk, Wazuh

Projects

Aegis | *Python, LangGraph, Ollama, ChromaDB, Docker* 🐙 karnayanarohith/Aegis (in active development)

- Engineering an autonomous, dual-mode Blue/Red Team security agent orchestrated with LangGraph and local LLMs via Ollama.
- Building a RAG-based memory pipeline using ChromaDB to store and recall threat intelligence, vulnerability logs, and attack patterns.
- Containerizing modular tools for automated network reconnaissance, log analysis, and incident triage mapped to the MITRE ATT&CK framework.

EXT4 Forensic Recovery Tool | *C, Linux, Systems Programming* 🐙 karnayanarohith/Aegis_File_Recovery_Tool

- Developed a low-level forensic recovery tool in C that parses raw EXT4 disk partitions directly without mounting the filesystem, recovering 13+ file types (including .pth, .h5, .pdf, and image formats) with 100% recovery accuracy in testing.
- Programmed manual superblock, inode table, and directory entry parsers to locate and reconstruct unlinked data block chains.
- Designed an ncurses-based TUI with a read-only preview engine to ensure forensically sound, non-destructive file recovery operations.

Experience

Android Security & Systems Research May 2026 – July 2026
Self-Directed Research, Realme C15 (MT6765)

- **CVE & Vulnerability Profiling:** Conducted MediaTek BROM exploit research to bypass Download Agent authentication using MTKClient and dump the device's seccfg partition. Mapped vulnerability attack surfaces on EOL firmware, profiling CVE-2022-20421 (Binder UAF) and CVE-2024-20106 (Type Confusion) via GDB and dmesg.
- **Custom ROM & Recovery Deployment:** Executed bootloader unlocking via BROM RAM patching and resolved Android dynamic partition allocation errors during custom recovery flashing. Deployed LineageOS and set up a Kali NetHunter chroot environment to convert the device into a mobile penetration testing platform.

Network Operations Lead March 2026
ELEXCENTRA Hackathon, Andhra University College of Engineering

- Engineered and managed the network infrastructure for a live, 5-hour hackathon supporting 400+ concurrent devices across 6 rooms.
- Configured and deployed Wi-Fi and physical LAN cabling using TP-Link switches along with D-Link and Netgear routers.

- Diagnosed and resolved real-time network congestion, signal degradation, and routing conflicts under pressure to ensure zero downtime.

Machine Learning Research Assistant

2025

NEXUS Research Project, Andhra University (under Dr. Karri Chiranjeevi)

- Engineered a real-time small object detection pipeline for industrial seals and tags utilizing PyTorch, CUDA, and the MMDetection framework.
- Benchmarked advanced CNN and Vision Transformer architectures including TOOD, Deformable DETR, and DINO.
- Achieved 98% accuracy on large objects and improved small-object mAP from 0.0 to 0.28 via resolution scaling and custom data balancing.

Achievements & Honors

1st Place Champion, HackAP (2025): Co-engineered 'Foundra' ( github.com/karnayanarohith/Foundra), an AI startup advisor orchestrating 11 autonomous LLM agents across 6 business development phases.

Security Solver (picoCTF, TryHackMe, OverTheWire): Top 5% (Bronze League) on TryHackMe, currently completing the SOC Level 1 learning path (20+ rooms resolved). Solving 'THE GAUNTLET' ( github.com/karnayanarohith/THE-GAUNTLET) — a 21-challenge security lab (2 completed, write-ups published on GitHub).

Project Finalist, Smart India Hackathon & AURORA 2.0 (2024): Developed computer vision models to analyze satellite imagery for illegal mining detection.

Certifications & Credentials

Centri Security — Blue Team Junior Analyst (BTJA)

Anthropic — Model Context Protocol (MCP) Developer

Kaggle — Machine Learning Certification

Cisco — Ethical Hacker path (18+ credentials), including: Application Vulnerability Exploitation; Post-Exploitation Techniques; Wireless Security

Deloitte — Cybersecurity Job Simulation (Log Analysis & Incident Response)

Tata Group — Identity & Access Management (IAM) Simulation

Languages

English (Professional), Telugu (Native)